EXPERIMENT NO: 3

TITLE: Design and Control any of DC-DC/ DC-AC/AC-DC/AC-AC Converters for Industrial Applications

AIM: To design and implement a DC-DC / DC-AC / AC-DC / AC-AC converter using MATLAB Simulink.

To analyze its performance and control characteristics under various load conditions.

To understand its industrial applications in power conversion and control systems.

APPARATUS:

Sr.No.	Equipment	Specification	Version
1	MATLAB IDE	Matlab Simulink model for circuit analysis	R2024a, R2025a

THEORY: Power electronic converters are essential components in modern industrial systems that enable efficient conversion and control of electrical energy. These static devices modify the form of electrical power—voltage, current, and frequency—to meet the specific requirements of various industrial loads. They play a vital role in ensuring precise power management, energy efficiency, and system reliability. Modern industries primarily employ four types of converters for controlled power conversion**:** DC–DC**,** DC–AC**,** AC–DC**,** and AC–AC converters.

DC-DC converters (choppers)→

DC-DC converters transform a DC voltage from one level to another. They use fast-switching techniques with semiconductor devices like MOSFETs or IGBTs to regulate the output voltage, providing a more efficient solution than older, linear regulators.

- The regulation of the output voltage is achieved by controlling the duty cycle, which is the ratio of the "on" time to the total switching period.

- The primary component is a power electronic switch, which is rapidly turned on and off to produce a pulsed DC voltage. This is then filtered using inductors and capacitors to create a smooth, steady DC output at the desired level.

- Common topologies include:

  Buck converter: Steps down the voltage.

  Boost converter: Steps up the voltage.

  Buck-boost converter: Can step voltage up or down.

  Cuk converters: It is a type of DC–DC switching converter that converts a positive input voltage into a regulated negative output voltage (or vice versa) using the principle of energy transfer through a capacitor. It combines the features of both buck and boost converters,

providing an output voltage magnitude that can be either higher or lower than the input voltage, with continuous input and output currents and low ripple characteristics.

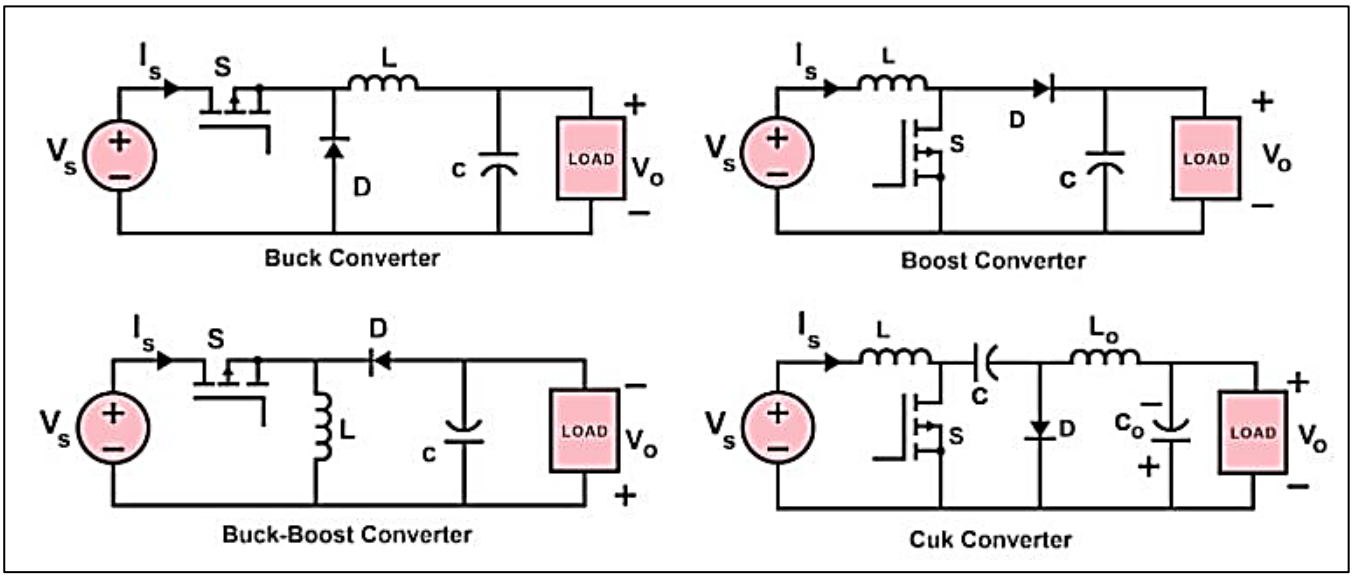
Industrial applications

- **Variable speed DC motor drives:** They control the motor speed by adjusting the DC voltage supplied to it.

- **Industrial automation:** Provide stable, low-voltage DC power (e.g., 24V) to sensors, actuators, and controllers from a main power supply.

- **Renewable energy systems:** Used to regulate the voltage from sources like solar panels or wind turbines before storage in batteries.

- **Electric vehicles (EVs):** Convert high-voltage battery power to the lower voltages required by the vehicle's auxiliary systems.



DC-AC converters (inverters)→

DC-AC converters, or inverters, convert DC power into AC power. This is essential for applications that use a DC source, such as batteries or solar panels, to drive standard AC loads.

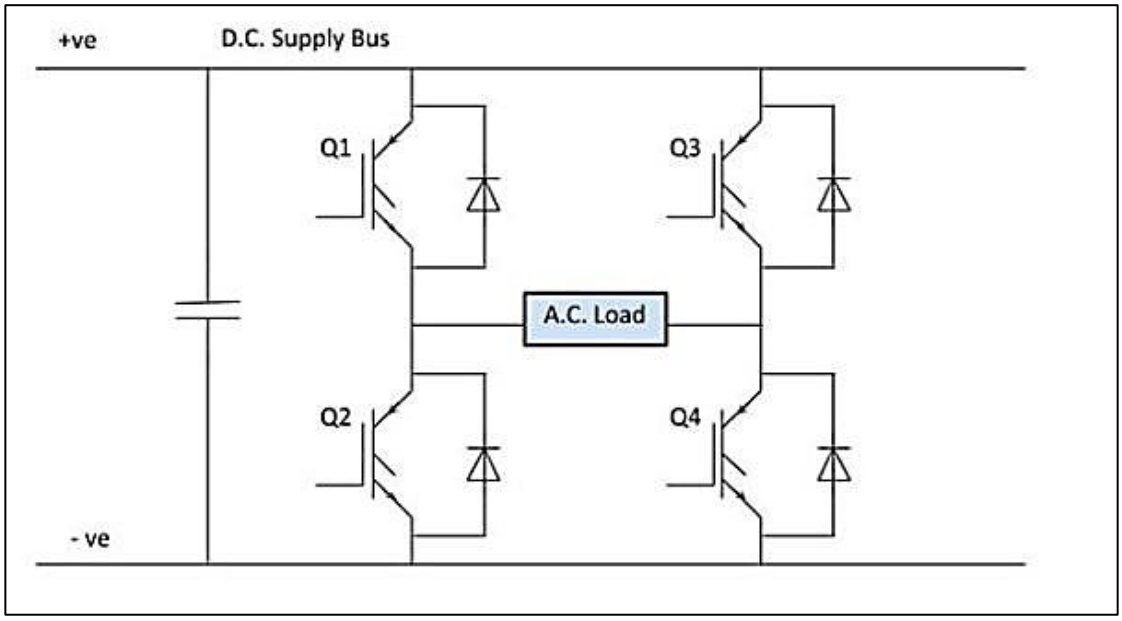
- The core of an inverter is an electronic switching circuit (an "inverter bridge") that reverses the polarity of the DC input to generate a square wave AC output.

- For higher quality output, inverters use techniques like Pulse Width Modulation (PWM) to shape the output into a pure sine wave, which is necessary for sensitive electronics.

- Output voltage and frequency are controlled by modulating the switching frequency and duty cycle of the semiconductor switches.

Industrial applications

- **Uninterruptible power supplies (UPS):** Use batteries (DC) and an inverter to provide continuous AC power to critical loads during a power outage.
- **Solar power systems:** Convert the DC output from photovoltaic (PV) panels into grid-compatible AC electricity.
- **Variable frequency drives (VFDs):** Inverter-based VFDs control the speed of industrial AC motors by adjusting the frequency and voltage of the power supplied.
- **Electric vehicles:** Used to convert the battery's DC power into the AC power needed to run the traction motor.



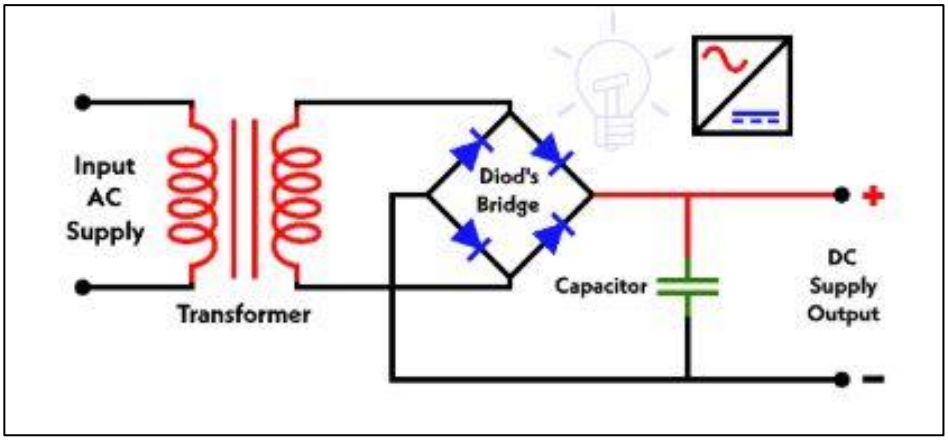
AC-DC converters (rectifiers)→

AC-DC converters, or rectifiers, convert AC power into DC power. This process is fundamental for powering most electronic equipment from the standard AC electrical grid.

- **Rectification:** Diodes or thyristors are used to convert the AC input into a pulsating DC waveform. For instance, a diode bridge rectifier can convert a full AC cycle into a series of positive voltage peaks.
- **Filtering:** An inductor-capacitor (LC) filter is used to smooth the pulsating DC into a stable, constant DC voltage.
- **Regulation:** A voltage regulator maintains the output at a constant level, protecting sensitive electronic equipment from voltage fluctuations.
- **Power factor correction (PFC):** More advanced rectifiers actively correct the power factor to minimize energy wastage and reduce harmonics introduced into the grid.

Industrial applications

- **Industrial motor drives:** Many industrial DC motors and automation systems require a DC power supply from the AC grid.
- **Telecommunications:** Network servers and switches rely on a stable DC power supply, often backed up by batteries, to ensure uninterrupted service.
- **Manufacturing:** Used to power the sensors, controllers, and electronic components in automated manufacturing systems.
- **Battery charging:** Large-scale industrial battery chargers for electric vehicles and forklifts use AC-DC converters.



AC-AC converters→

AC-AC converters convert AC power at one voltage or frequency to AC power at another voltage or frequency. Unlike simple transformers, they use semiconductor switches to provide precise control over the output.

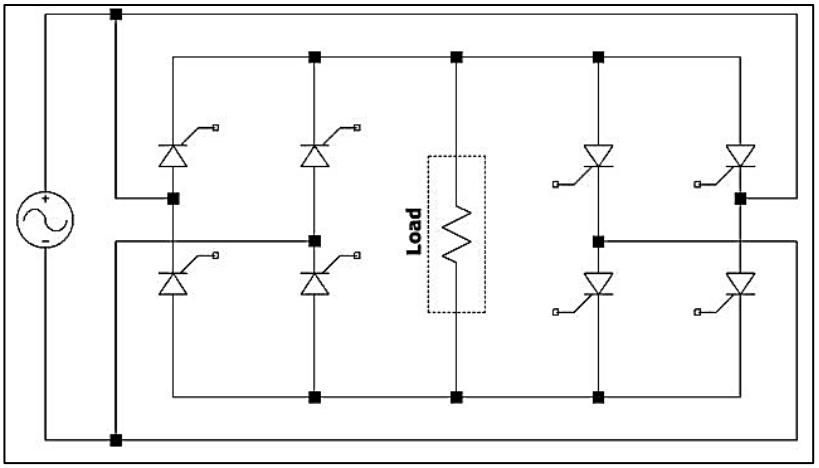
- **Cycloconverters:** Directly convert AC power of one frequency to AC power of a lower frequency. They are used for high-power, low-speed motor drives.
- **Matrix converters:** Use a direct connection of bidirectional switches to link the input phases to the output phases without a bulky DC link capacitor. This design results in a more compact and efficient converter.
- **AC voltage controllers:** Use thyristors or TRIACs to control the RMS output voltage by adjusting the firing angle, suitable for applications like heating and lighting control.

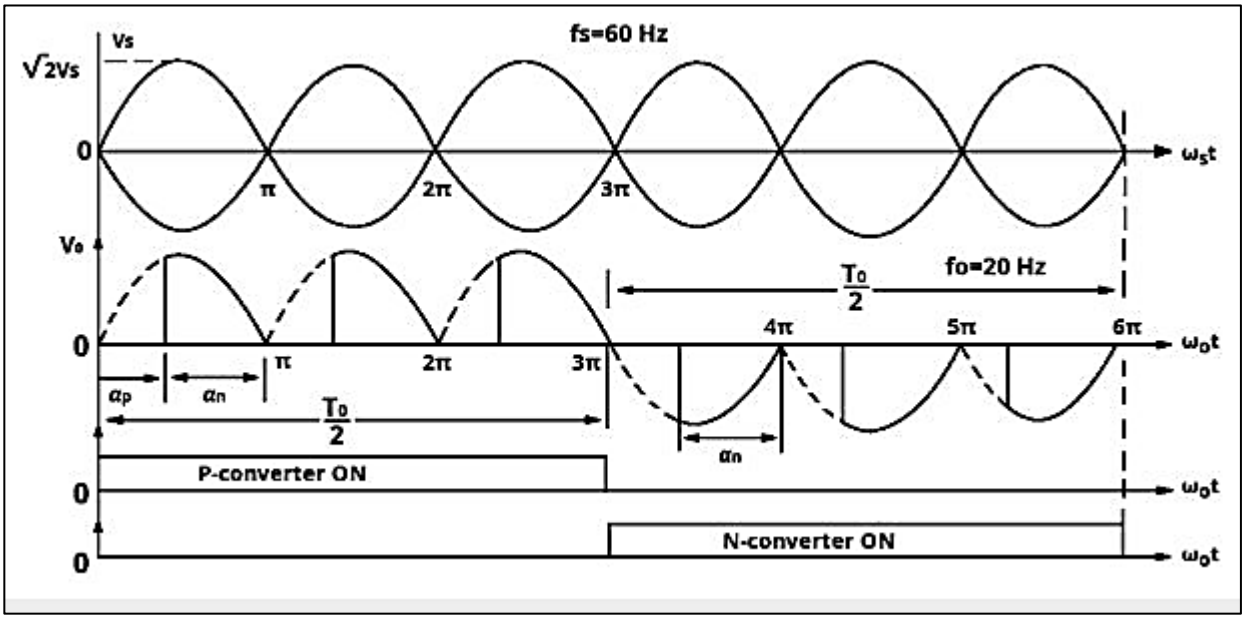
Industrial applications

- **Variable frequency motor drives:** Used to control the speed and torque of large induction motors, which are the main workhorses of heavy industry.
- **High-power applications:** Matrix converters and multilevel converters are used in high-power systems to manage voltage and frequency with improved power quality.

- **Industrial heating and lighting:** Voltage controllers provide simple, effective control over power to resistive loads.

- **Railway transportation:** AC-AC converters enable regenerative braking and power control in electric locomotives.





RESULT:

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CONCLUSION:

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